Paleomagnetism of the upper Keweenawan sediments: the Nonesuch Shale and Freda Sandstone

STEVEN G. HENRY, FREDERICK J. MAUK, AND ROB VAN DER VOO

Department of Geology and Mineralogy, The University of Michigan, Ann Arbor, MI, U.S.A. 48109

Received 23 July 1976

Revision accepted for publication 30 November 1976

The natural remanent magnetization of the upper Keweenawan Nonesuch Shale and Freda Sandstone has been analyzed with thermal, alternating field, and chemical demagnetization techniques. The results of this study are in good agreement with previously published works by DuBois and by Vincenz and Yaskawa, but place a tighter constraint on the North American apparent polar wander path. Fifty-eight samples, representing nearly 900 m of section, have been collected from the flanks of the Porcupine Mountain uplift. From principally thermal demagnetization analyses, a mean direction of primary magnetization has been calculated for the Nonesuch Shale, with declination 279.8°, inclination + 9.8°, yielding a virtual geomagnetic pole position at 176.5° E, 10.3° N, and for the Freda Sandstone, with declination 271.3°, inclination + 0.7°, yielding a virtual geomagnetic pole at 179.5° E, 1.2° N. A group of intermediate (secondary) components of magnetization is removed between temperatures of 350°C and 550°C, yielding well clustered directions. Its mean direction with declination 280.6°, inclination −9.5°, resulted in a virtual geomagnetic pole at 169.2° E, 3.7° N. This secondary magnetization is assumed to be of chemical origin and is most likely associated with the late Precambrian copper mineralization of the Nonesuch Shale. By thorough sampling of the stratigraphic column it is possible to infer the general direction of motion of a plate as the sediments were deposited. The motion of the North American plate as observed in the upper Keweenawan magnetizations is in agreement with the previously published polar wander paths for the late Precambrian.

On a analysé la magnétisation rémanente naturelle du schiste argileux de Nonesuch et du grès de Freda du Keweenawien supérieur en utilisant des techniques de démagnétisation thermiques, avec champ alternatif et chimiques. Les résultats de cette étude s'accordent bien avec les travaux publiés par DuBois et par Vincenz et Yaskawa, mais posent une contrainte plus forte sur le tracé apparent du déplacement des pôles en Amérique du Nord. On a prélevé 58 échantillons, représentant environ 900 m de section dans les flancs du soulèvement de Porcupine Mountain. A partir des analyses de démagnétisation thermique surtout, on a calculé une direction principale de magnétisation primaire pour le schiste argileux de Nonesuch, avec une déclinaison de 279.8° et une inclinaison de +9.8°, ce qui donne comme position du pôle géomagnétique virtuel, 176.5° E et 10.3° N; pour le grès de Freda, avec une déclinaison de 271.3° et une inclinaison de +0.7°, on obtient la position du pôle géomagnétique virtuel à 179.5° E et 1.2° N. Un groupe de composantes intermédiaires (secondaires) de magnétisation disparaissent entre les températures de 350°C et 550°C, résultant en un faisceau de directions. Leur direction moyenne, avec une déclinaison de 280.6° et une inclinaison de −9.5° situe le pôle géomagnétique virtuel à 169.2° E et 3.7° N. On suppose que la magnétisation secondaire est d'origine chimique et est probablement associée à la minéralisation de cuivre dans le schiste argileux de Nonesuch à la fin du Précambrien. En échantillonnant soigneusement tout le profil stratigraphique, il est possible de déduire la direction générale du mouvement d'une plaque au cours du dépôt des sédiments. Le mouvement de la plaque nord-américaine telle qu'observée par les magnétisations du Keweenawien supérieur s'accorde avec les données publiées antérieurement sur la migration des pôles à la fin du Précambrien.

[Traduit par le journal]

Can. J. Earth Sci., **14**, 1128–1138 (1977)

Introduction

This study has been undertaken to reexamine the paleomagnetism of the upper Keweenawan sedimentary rocks, the Freda Sandstone, and the Nonesuch Shale. These sedimentary rocks occur in the upper peninsula of Michigan, part of the North American craton since their deposition about 1050 million years ago. Using modern demagnetization techniques it is possible to obtain reliable magnetic directions from rocks of this age and older, and determination of paleomagnetic poles for these ages constrains the North American polar wander path (PWP) in its extension into the Precambrian. Once considerable parts of the PWP are well determined, and when contemporaneous poles from different

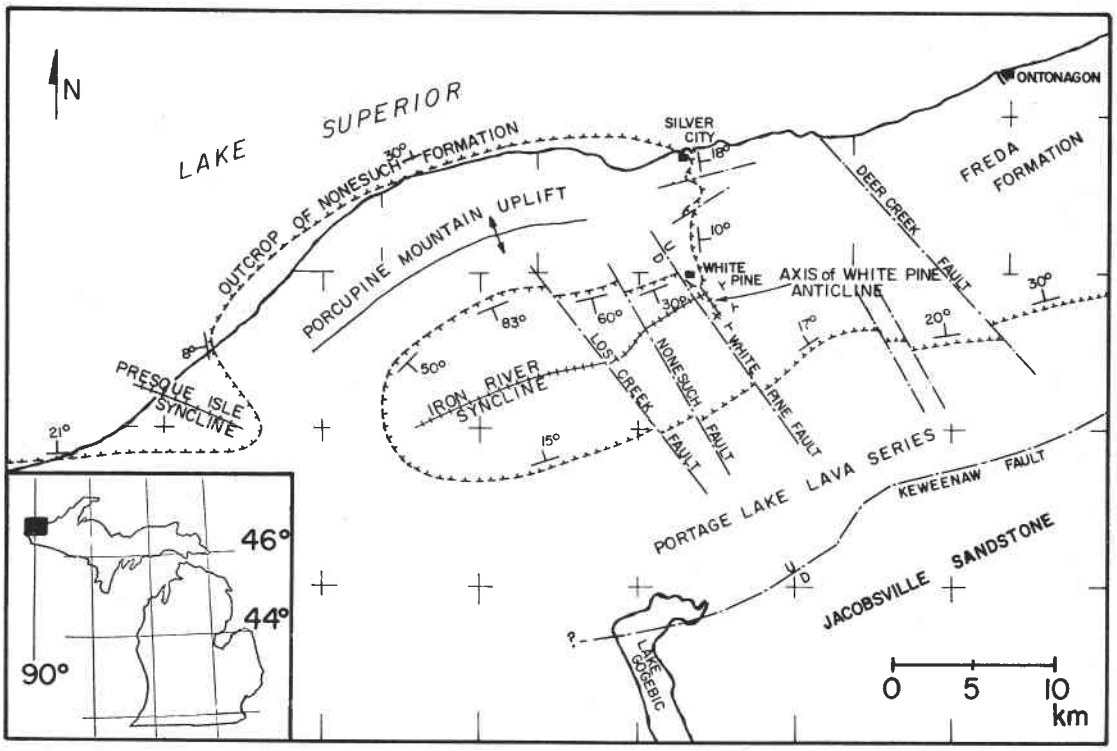


FIG. 1. Schematic tectonic map of the Porcupine Mountain uplift and surrounding area. Inset gives the location in the upper peninsula of Michigan.

areas, including the orogenic belts, are accurately established, poles lying off the path may be unraveled. In this regard, the ages of the Nonesuch Shale and Freda Sandstone (about 1050–1060 Ma) are important, since it is between 1125 and 1000 million years ago that the major pre-collision movement of part of the Grenville Province is thought to have occurred with respect to interior Laurentia (Irving and McGlynn 1976, Fig. 18).

Geology and Sampling

In selecting the sampling sites, care was taken to avoid areas where rotation may have been introduced by folding or faulting. Some of these faults can be seen offsetting the sedimentary rocks surrounding the Porcupine Mountain uplift (Fig. 1). The Keweenaw fault is a high-angle thrust fault which traverses the length of the Keweenaw Peninsula, bringing the upper Keweenawan sediments and middle Keweenawan lavas in contact with the Jacobsville Sandstone, a late Precambrian (Cambrian?) formation. The westernmost extent of the Keweenaw Fault is shown in Fig. 1, and it is unlikely that the sampling areas have been affected by movement along this fault.

A total of 58 samples (yielding 127 specimens) were collected at 31 sites in two separate areas. Twenty-five of the sites are located on the eastern end of the Porcupine Mountain uplift, and the remaining six are located in the Presque Isle Syncline at the western end (Fig. 2). Excellent outcrops are found in both areas, with good exposures at the Lake Superior shoreline and in the beds of the numerous rivers which drain into the lake. The Porcupine Mountain uplift originated through the deformation of the middle Keweenawan sediments and volcanics and the upper Keweenawan sediments into a series of gently undulating synclines and anticlines. Where sampled, the strata have dips of less than 20°. Near the southern margin of the uplift, the dips are as high as 83°, but outcrops here were less accessible.

The upper Keweenawan sediments conformably overlie the middle Keweenawan sediments and Portage Lake Lavas. These lavas are basaltic

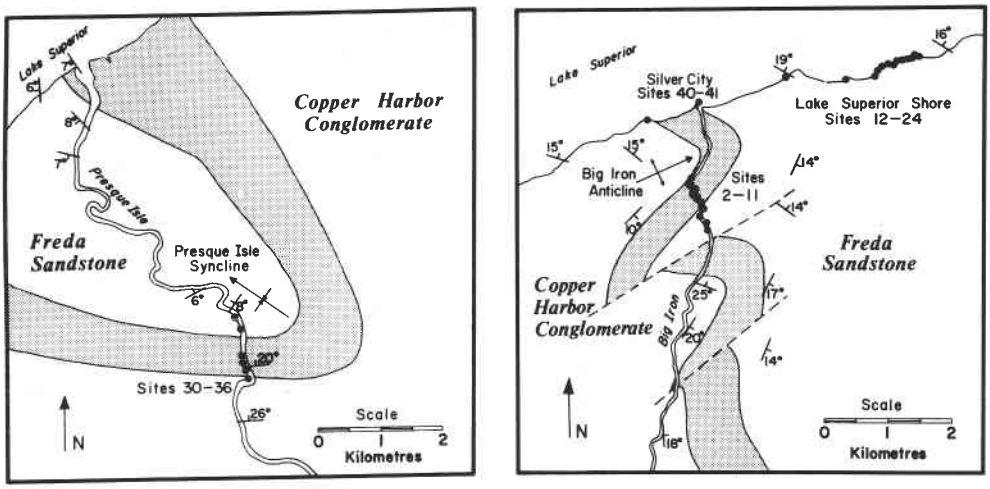


FIG. 2. Geological maps of the two sampling areas on either side of the Porcupine Mountain uplift. Sampling sites are indicated by dots. The Nonesuch Shale is shown in stippling.

and andesitic in composition and are presumed to be associated in time of emplacement with the Duluth Gabbro. The Duluth Gabbro has been dated by Faure *et al.* (1969) and yields a K–Ar age of 1115 ± 15 Ma. The age most generally assigned to the Portage Lake Lavas is a K–Ar age of 1100 Ma (*e.g.* Irving and McGlynn 1976).

The upper Keweenawan sedimentary rocks, the Freda Sandstone, and the Nonesuch Shale overlie the middle Keweenawan Copper Harbor Conglomerate and the Portage Lake Lavas. The Copper Harbor Conglomerate conformably overlies the Portage Lake Lavas and consists of a matrix with boulders and pebbles mainly of rhyolitic composition. Beds of coarse red sandstone are intercalated. Within the unit there are three major rhyolitic flows concentrated mainly at the base of the formation. However, one flow occurs near the top, close to the base of the Nonesuch. The rhyolitic flows have not been dated radiometrically, but they are most likely associated in time with the Mellon granite, located in Wisconsin, about 10 km to the southwest. Chaudhuri (S. Chaudhuri, personal communication, 1976) has been dating the Keweenawan igneous activity. Using biotite extracted from the Mellon granite, a Rb–Sr age of 1075 Ma is obtained. Rhyolitic flows in the Porcupine Mountain State Park and a flow near Bergland, about 2 km southeast of the park, yield similar Rb–Sr ages ranging between 1040 and 1088 Ma. Chaudhuri currently is trying to determine whether these ages may have been reset and whether the igneous activity actually could be older, perhaps 1100–1200 Ma.

The Nonesuch Shale consists of prodeltaic sediments, rich in organic matter in many areas. The contact between the shale and the Copper Harbor Conglomerate is extremely variable, ranging from a disconformable contact to a gradational conformity. The Nonesuch is thinly bedded at its base, grading upward into siltstone and finally into a fine-grained banded sandstone, known as the 'tiger stone', which marks the base of the Freda. The Nonesuch is fairly constant in thickness, ranging from 150–200 m. The Freda is a fine- to medium-grained sandstone interbedded with numerous red siltstones and shales. It overlies the Nonesuch and is much more extensive, representing nearly 3500 m of section in the White Pine district. Numerous sedimentary structures, including climbing ripples and extensive cross bedding of the fine-grained sandstone, indicate rapid deposition in a deltaic environment. The orientation of the sedimentary structures suggests that the source area for both the Nonesuch Shale and the Freda Sandstone was to the south.

The upper Keweenawan sedimentary rocks extend from the Keweenaw Peninsula into Wisconsin. The Freda Sandstone of the Keweenaw Peninsula is intruded by the Bear Lake Rhyolite. This is the only known intrusive into the upper Keweenawan rocks. The rhyolite has been dated by Chaudhuri (S. Chaudhuri, personal communication, 1976), and yields a Rb–Sr age

of 1007 ± 25 Ma. This sets the minimum age for the sediments. The maximum age is determined by the Portage Lake Lavas, and sets an upper limit of approximately 1100 Ma. Previously published ages for the Nonesuch Shale (Barghoorn *et al.* 1965; Vincenz and Yaskawa 1968) are within these limits. This brackets the age of the upper Keweenawan sediments between about 1000 Ma and 1100 Ma. We have assigned representative ages of 1060 Ma for the Nonesuch Shale, and 1050 Ma for the Freda Sandstone. The previous age is 1046 ± 46 Ma is based on Rb–Sr ratios of whole rock samples from the Nonesuch of the White Pine area, as determined by Chaudhuri, and was confirmed by him as still valid.

Measurement and Demagnetization

Measurement of the natural remanent magnetization (NRM) was carried out with a Schonstedt spinner magnetometer. The samples from the Freda Sandstone ranged in intensity between 0.03 A m^{-1} (0.3×10^{-4} emu/cm^3) and 0.1 A m^{-1} with an average of 0.08 A m^{-1}. The intensities of the Nonesuch samples were generally lower, ranging between 0.01 and 0.08 A m^{-1} with an average value of 0.02 A m^{-1}.

Mineralogical investigations by Vincenz and Yaskawa (1968), who studied both thin sections and polished sections, indicate that the main magnetic minerals are hematite or titanohematite. The Curie temperature measurements of 640°C and 675°C also correspond to titanohematite and hematite. From the response of both the Nonesuch and Freda to thermal demagnetization, and the lack of response to alternating field (AF) demagnetization, we conclude that our study is in agreement with the mineralogical findings of Vincenz and Yaskawa.

All samples from the 31 sites were cleaned using either chemical or thermal demagnetization. Alternating fields failed to remove more than 10% of the NRM in samples of the Freda, though in some samples of the Nonesuch as much as 60% could be removed by the use of 100 mT (1000 Oe). However, due to the generally poor response to AF demagnetization, cleaning was completed using mainly thermal demagnetization. Generally, components of secondary magnetization were removed by temperatures of about 550°C, whereupon the magnetization decayed univectorially to the origin. Progressive demagnetization diagrams are shown in Figs. 3 and 4, plotted according to the Zijderveld (1967) method. The characteristic directions thus obtained for the Nonesuch Shale and the Freda Sandstone are shown in Fig. 5.

Consistent orientations of a secondary component of magnetization may be observed in Fig. 3. These components were eliminated by temperatures between 350°C and 550°C. We assume that in this temperature range only the intermediate (secondary) component is being removed, since the trajectories are straight and often defined by three or more successive measurement points (Fig. 3) and since in those samples, which do not show this intermediate component, appreciable decay of the magnetic vector does not begin until above temperatures of 550°C (*e.g.* Fig. 4, sample 26). Vector subtraction was used to determine the directions of these components, but only for those samples which showed a distinct multivectorial behavior such as that shown in Fig. 3. The method used for this vector subtraction will not be described

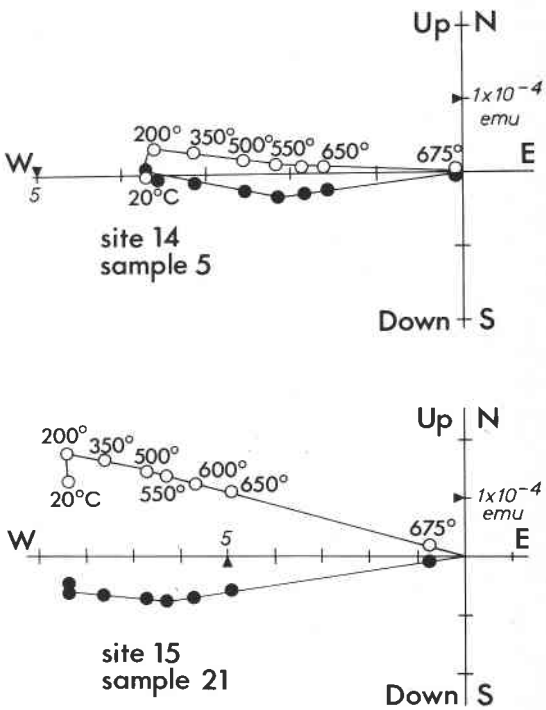


FIG. 3. Orthogonal projections (demagnetization vector diagrams) of the successive endpoints of the magnetization vector during progressive thermal demagnetization. Solid symbols represent projections on the horizontal plane, open symbols on the east–west vertical plane. Note the change in direction at 550°C, below which a slightly different, secondary magnetization is being eliminated.

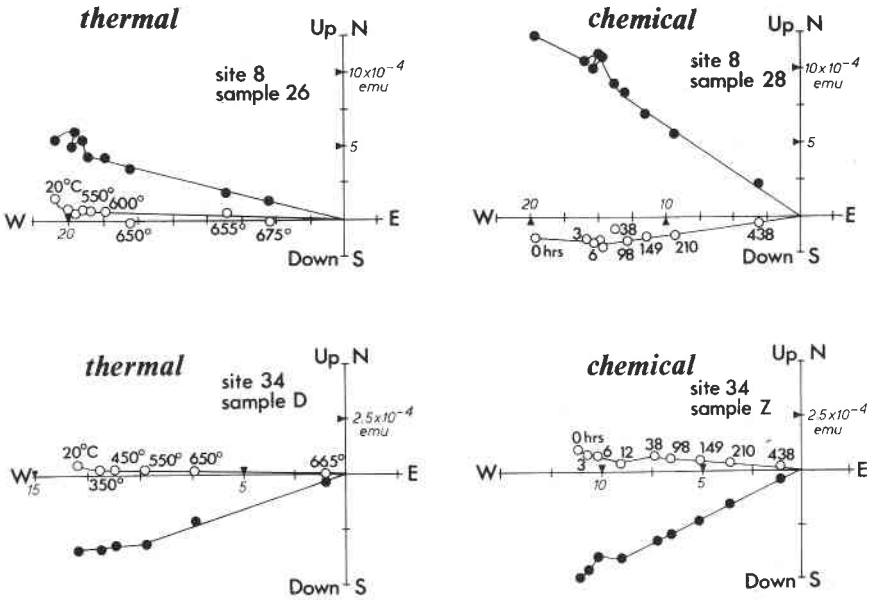


FIG. 4. Chemical and thermal demagnetization vector diagrams. The symbols and conventions are the same as those used in Fig. 3.

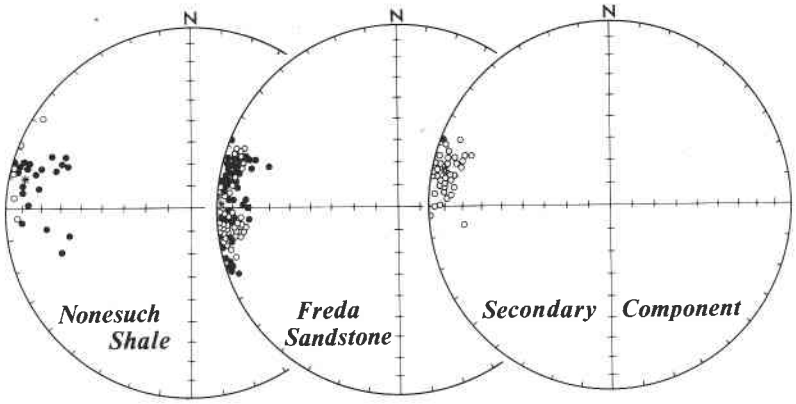


FIG. 5. Equal area projections of the characteristic and intermediate (secondary) directions of magnetization. Open symbols pointing upward, solid symbols pointing downward.

here since it is extensively discussed by Zijderveld (1967, 1975). The intermediate directions were observed in 36 specimens, and they revealed a cluster which is very close to the directions of the higher-temperature, characteristic magnetizations. Since, however, the directions of the secondary magnetizations were quite discreet (Fig. 3), we will discuss their significance below. These directions are also plotted in Fig. 5.

Chemical demagnetization was performed on 12 samples. The samples were soaked at set periods of time in 10 *N* HCl (Park 1970). The chemical demagnetizations (even though carried out in the earth's field) are in excellent agreement with the thermal demagnetizations performed in field-free space. A comparison of chemical and thermal demagnetizations is illustrated in Fig. 4.

The ratio of ferric (Fe^{3+}) to ferrous (Fe^{2+}) ions was determined on a sample which displayed a strong secondary magnetization. The total iron content was determined to be 4.2 wt.% using atomic absorption. Using the Wilson method of ferrous iron determination, less than 0.4 wt.% Fe^{2+} was found. This result supports the

TABLE 1. Paleomagnetic summary of the site-mean directions

Site	Position	R_{sp}	k	α_{95}	Decl. (deg)	Incl. (deg)	Pole position		Smoothed average	
							Long. (E)	Lat.	Long. (E)	Lat.
17	667	4/4	319	5.1	286.5	−8.9	165.3°	7.8° N	—	—
22	645	2/2	1322	6.9	254.8	−5.8	189.7°	12.5° S	—	—
23	628	2/2	2495	5.0	272.1	+16.5	184.3°	7.6° N	180.4°	3.4° N
24	615	4/4	198	6.5	266.9	+13.1	187.3°	2.8° N	183.5°	0.1° N
16	612	10/6	569	2.8	281.2	+8.8	175.4°	10.9° N	181.9°	2.1° N
21	594	4/4	1244	2.6	264.2	−11.9	180.6°	8.3° S	181.9°	1.1° S
20	579	4/4	80	10.3	267.2	−2.2	181.8°	2.7° S	180.8°	1.3° S
15	557	9/9	143	4.3	261.7	−6.8	184.3°	8.2° S	182.3°	4.9° S
14	535	10/9	176	3.9	267.2	−0.1	181.8°	2.0° N	182.1°	3.7° S
13	515	4/4	261	5.7	263.1	−7.5	182.9°	7.4° S	181.3°	2.9° S
12	486	3/3	270	7.5	269.1	−3.8	179.9°	2.0° S	177.2°	0.7° N
18	407	6/5	90	8.1	273.1	−2.3	177.4°	1.3° N	175.9°	2.7° N
19	271	4/4	787	3.3	288.7	−8.4	163.8°	9.5° N	176.7°	3.6° N
6	84	3/3	90	13.1	282.0	+10.0	175.1°	11.9° N	175.0°	6.5° N
35	73	3/3	721	4.6	263.1	+5.5	187.0°	2.7° S	178.3°	8.8° N
5	52	4/3	97	12.6	285.0	+5.6	171.3°	12.2° N	179.7°	9.4° N
34	28	7/6	877	2.3	250.9	+1.6	194.5°	12.4° N	179.6°	10.8° N
4	21	4/4	2065	2.0	285.5	+4.9	170.4°	12.4° N	176.9°	12.9° N
40	15	4/4	278	5.5	287.4	+19.0	174.4°	19.0° N	178.4°	12.8° N
3	9	2/2	2558	4.9	279.9	+2.8	174.2°	7.8° N	175.6°	9.2° N
					Freda–Nonesuch boundary					
2	−17	3/3	275	7.4	279.3	+14.0	178.5°	11.6° N	177.5°	8.4° N
7	−47	4/1	—	—	266.6	−7.3	180.4°	5.0° S	175.8°	6.8° N
8	−65	5/5	10	25.7	276.0	+11.7	180.1°	8.4° N	176.1°	9.8° N
9	−81	4/3	13	35.3	288.0	−3.6	165.9°	10.9° N	179.3°	8.1° N
33	−100	4/4	301	5.3	289.4	+25.6	175.5°	23.0° N	179.0°	10.6° N
10	−122	1/1	—	—	261.9	+21.7	194.0°	2.7° N	177.5°	10.2° N
32	−137	4/1	—	—	275.6	+9.3	179.2°	7.2° N	178.5°	9.4° N
11	−138	3/3	136	10.6	280.0	−0.6	172.9°	6.6° N	180.7°	4.8° N
31	−167	1/1	—	—	281.6	−2.2	170.8°	7.2° N	177.0°	6.6° N
30	−183	1/1	—	—	265.2	+9.0	186.7°	0.1° N	—	—
41	−200	4/4	136	7.9	281.5	+10.2	175.5°	11.6° N	—	—

NOTE: Position—in meters positive above Freda–Nonesuch contact, negative below contact; R_{sp} is the ratio of specimens measured to specimens used for the statistical analysis; k and α_{95} are statistical parameters (Fisher 1953); Decl. = declination and Incl. = inclination; the smoothed average is calculated from five mean poles (two on each side), weighted equally.

mineralogical findings of Vincenz and Yaskawa (1968), and eliminates magnetite as a carrier of either the primary or secondary components of magnetization.

Five additional specimens were drilled from the hand samples on which the ferric–ferrous determination was made. These samples were chemically demagnetized to determine when the secondary component was removed. A sample was sectioned after each 24 h treatment to reveal the results of the chemical leaching. The cross sections revealed that the red pigmentation was being removed uniformly radially from the surfaces in contact with the acid. The secondary component was removed after 72 h of leaching. Most of the pigmentation remained, however, indicating that the pigmentation was carrying not only the secondary magnetization, but both components of magnetization. The continued leaching produced demagnetization diagrams like those in Fig. 4, with a univectoral decay to the origin from 72–450 h.

Results

The sampling at both sides of the Porcupine Mountain uplift provided a good stratigraphic coverage of nearly 900 m of section. Site density was a function of suitable outcrops, with the less dense coverage being in the very thinly bedded shales and siltstones. The results from the 31 sites were tabulated and presented in Table 1. The sites are listed with the uppermost site of the section at the top of the table. Their position in the section is given in metres from the Freda–

TABLE 2. Mean paleomagnetic results before and after correction for the tilt of the strata

	Before tilt correction							After tilt correction					
						Pole position						Pole position	
	R_s	Decl. (deg)	Incl. (deg)	k	α_{95}	Long. (E)	Lat.	Decl. (deg)	Incl. (deg)	k	α_{95}	Long. (E)	Lat.
Nonesuch, giving unit weight to site means	11/11	275.0	−0.7	21	10.2	176.4°	3.2° N	276.8	+8.1	35	7.8	178.1°	7.6° N
Nonesuch, giving unit weight to specimens	34/27	277.5	+0.2	14	7.7	174.8°	5.2° N	279.8	+9.8	22	6.1	176.5°	10.3° N
Freda, giving unit weight to site means	20/20	272.7	−7.0	27	6.4	175.9°	0.7° S	272.4	+1.5	32	5.9	179.0°	2.2° N
Freda, giving unit weight to specimens	93/85	271.8	−7.6	27	3.0	176.3°	1.6° S	271.3	+0.7	31	2.8	179.5°	1.2° N
Secondary component (specimens)	127/34	282.6	−16.6	59	3.2	165.3°	2.3° N	280.6	−9.5	81	2.7	169.2°	3.7° N

NOTE: R_{sp} is the ratio of sites (specimens) measured to sites (specimens) used in the statistical analysis. Other symbols are the same as those explained in Table 1.

Nonesuch contact, upwards into the Freda being positive, and downwards into the Nonesuch being negative. The ratio of the number of samples measured to the number (N) of samples used in the computation of the mean pole is given by R_{sp}; k is an estimate of the precision parameter and α_{95} is the cone of confidence at the 95% probability level (Fisher 1953). The mean declinations and inclinations are listed after restoring the strata to the paleohorizontal. All poles presented in this paper are assumed to be north poles; all directions thus represent normal polarity. No identifiable reversed polarity directions were observed in the samples. The final column in the table consists of running-average poles. These values are computed as mean poles, using five entries (with two on each side), all of which are weighted equally. This running average is intended to smooth secular variation and experimental inaccuracies.

The fold test was performed by computing the mean pole both before and after restoring the strata to the paleohorizontal. The dips involved are small, less than 20° at all sites; however, an improvement in the statistical parameters is observed upon unfolding, demonstrated in Table 2. Both the characteristic and intermediate secondary magnetizations apparently were acquired before folding.

The mean direction of characteristic magnetization for the Freda Sandstone and the Nonesuch Shale, as well as their virtual geomagnetic pole positions are given in Table 3. There is no reason to doubt that these magnetizations represent the original (primary) remanence of the beds, particularly since they show a convincing pattern of secular variation superimposed upon a gradual apparent polar wander trend. The distinct secondary (intermediate) magnetization acquired by both the Freda and the Nonesuch yields a virtual geomagnetic pole position at 169.2° E, 3.7° N which differs by about 10° from the poles of the primary magnetization. The statistical parameters of all of these results are presented in Table 2 and summarized in Table 3. The resulting pole positions and other poles of late Keweenawan age are plotted in Fig. 6; their relationships will be discussed later.

The Freda and Nonesuch were previously sampled by DuBois (1962) and by Vincenz and Yaskawa (1968). In these publications the Freda and Nonesuch were combined to yield one pole.

TABLE 3. Summary of paleomagnetic results from the upper Keweenawan sediments

Author	N	k	α_{95}	Decl. (deg)	Incl. (deg)	Pole position Long. (E)	Pole position Lat. (N)	dp	dm
DuBois (1962): Freda and Nonesuch	68	26	3	285.0	−1.0	169.0°	9.0°	1.5	3.0
Vincenz and Yaskawa (1968): Freda and Nonesuch	5	31	14	279.0	+2.0	176.0°	7.0°	7.0	14.0
Vincenz and Yaskawa (1968): Nonesuch–White Pine	3	17	31	279.0	+15.0	179.0°	11.0°	16.3	31.6
This study: Nonesuch	27	22	6	279.8	+9.8	176.5°	10.3°	3.1	6.2
This study: Freda	85	31	3	271.3	+0.7	179.5°	1.2°	1.4	2.8
This study: secondary component	34	81	3	280.6	−9.5	169.2°	3.7°	1.4	2.7

Their results are tabulated with the results of this study in Table 3. The slight discrepancy between the pole determined by DuBois and the present results is most likely due to the more effective removal of secondary components of magnetization in this study. DuBois demagnetized two samples thermally. Because he observed only small changes in the direction of magnetization upon treatment, NRM results of his remaining 66 samples were deemed satisfactory.

Our results are in close agreement with the results by Vincenz and Yaskawa (1968). The majority of the samples on which their pole is based were collected from the Keweenaw Peninsula. In their publication they have listed separately the results from the White Pine mine, and tabulated this pole as anomalous. This 'anomaly', however, is most likely due to a real change in the pole position. Their results were based on only three samples, all of which were from the base of the Nonesuch. Their values from the base of the Nonesuch, as indicated in Table 3, are in excellent agreement with the values of this report from the base of the Nonesuch, as can be seen in Table 1 (site 41). The results from this study indicate that it is possible to detect changes in the pole position as one moves down the section. This is seen in the changes of the smoothed averages in the last column of Table 1; the trend appears to be an increase in distance to the pole with decreasing age. The change in inclination, together with the lack of change in the declination, agrees with the general apparent PWP of the North American continent as described for 1150–1000 Ma by Spall (1971), Elston and Scott (1973), Irving *et al.* (1974), and Irving and McGlynn (1976).

The pole positions determined from this study agree very well with the previously published poles for rocks of late Keweenawan age (1100–1000 Ma). The poles from this period (taken from Irving and McGlynn 1976; Figs. 5, 7), as well as the results from this report (triangles and square) and a pole for the Jacobsville Sandstone from the upper peninsula of Michigan (French 1976), are tabulated and plotted in Fig. 6. The poles for the Freda and Nonesuch lie within the cluster, fitting well with the poles between 1000 and 1060 Ma.

Discussion of the Secondary Magnetization

Our demagnetization results clearly indicate two separate periods of magnetization acquisition in both the Nonesuch Shale and the Freda Sandstone. The primary magnetizatio was

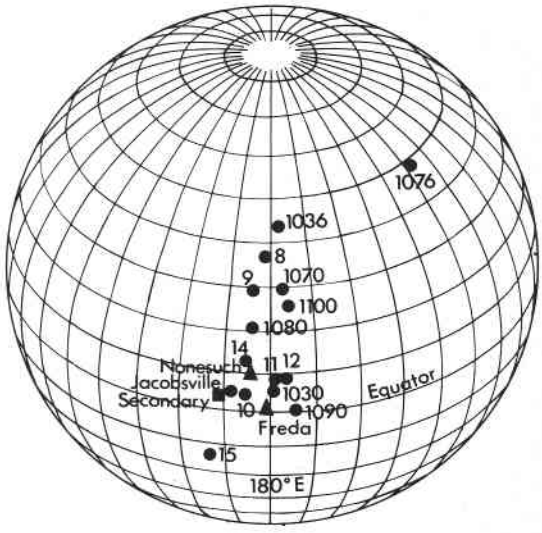


FIG. 6. Late Keweenawan pole positions for North America. The poles for the Nonesuch Shale, the Freda Sandstone (triangles) and for the secondary magnetization (square) are from this study. The pole labeled Jacobsville is from French (1976) and is tabulated with other poles in Table 4.

TABLE 4. Pole positions taken from French (1976), Irving and McGlynn (1976, Fig. 5; identified here by approximate age indication), and Elston and Grommé (quoted in Irving and McGlynn 1976, Fig. 7; identified here by two-digit numbers which increase with decreasing age)

Approximate age (Ma)	Formation	Identification number	Pole position
650?	Jacobsville	—	5.0° N, 171.0° E
995	Nankoweap Fm, upper	15	14.2° S, 166.9° E
1005	Nankoweap Fm, ferruginous	14	12.5° N, 174.4° E
1030	Pikes Peak granite	1030	6.0° N, 181.0° E
1036	Nemegosenda carbonatite	1036	43.0° N, 182.0° E
1050	Freda Sandstone	—	1.2° N, 179.5° E
1060	Nonesuch Shale	—	10.3° N, 176.5° E
1070	Mamainse Lavas, normal	1070	29.5° N, 182.7° E
1976	Mamainse Lavas, reversed	1076	48.5° N, 232.0° E
1080	Copper Harbor	1080	21.0° N, 176.0° E
1085	Sandstone with Cardenas	12	8.6° N, 183.7° E
1090	Cardenas Lava Flows	11	8.3° N, 181.5° E
1090	Cardenas Lavas	1090	0.4° N, 185.4° E
1092?	Dox Sandstone, upper	10	4.6° N, 174.4° E
1094?	Dox Sandstone, upper–middle	9	29.2° N, 175.3° E
1096?	Dox Sandstone, middle	8	37.1° N, 178.3° E
1100	Portage Lake Lavas	1100	25.2° N, 184.0° E

probably acquired during or shortly after deposition. As deposition continued, the motion of the North American continent was recorded in the magnetizations. The smoothed average reflects this motion as a change in inclination with only a slight change in declination, corresponding to plate motion away from the north pole.

The secondary magnetization must have been acquired later in the history of the sediments, but how much later is not easy to determine. Vincenz and Yaskawa (1968) observed a directional change in the samples from the White Pine mine when they were heated to temperatures above 500°C. However, at 530°C their samples showed directional instability, believed to be due to the approaching of the blocking temperatures. This prevented further examination of the magnetizations. They believed the directional change to be due to a removal of a viscous partial thermoremanent magnetization, imposed at the time of copper mineralization. From their observed directional change during demagnetization they felt that the secondary magnetization was acquired during late Precambrian or Early Cambrian time.

The pole as determined from the secondary magnetization in our samples indicates that the secondary magnetization was indeed acquired during Cambrian or late Precambrian times. It must be remarked that this pole is close to the late Keweenawan poles, but lies more to the west in the vicinity of Cambrian and late Hadrynian poles (Van der Voo *et al.* 1976). We believe, for reasons given below, that the secondary magnetization was acquired chemically rather than thermally, although we agree that it is probably associated with the copper mineralization process.

Evidence for a continued low-temperature environment from the time of deposition to the present is available from two sources. The Nonesuch Shale is very rich in organic materials. Barghoorn *et al.* (1965) studied the organic material for evidence of Precambrian life forms. The organic material which they identified could not survive elevated temperatures. They argue, for example, that the porphyrins could persist no longer than 100 years at a temperature of 250°C, 11.5 days at 350°C, and only 100 s at 550°C. The copper mineralization is believed to have taken approximately 1 million years (Brown 1971), so for the porphyrins to be preserved a temperature much below 250°C is necessary.

The second line of evidence is more restrictive and is based on low-temperature minerals identified in the Nonesuch Shale. Brown (1971) studied the process of copper mineralization in the White Pine mine. His identification of the

mineral djurleite suggests a maximum temperature of 93°C for the shales, and the existence of a pink bornite indicates a maximum temperature of less than 75°C. These two lines of evidence point to a maximum temperature of less than 100°C.

Brown (1971) reviews the finding of White (1966), which offers as a mechanism for the copper mineralization, a fluid migration involving ionic exchange between the more saline solutions below the Nonesuch Shale in the Copper Harbor, with the less saline solutions above the Nonesuch in the Freda Sandstone. The migration of ions concentrated the copper in the Nonesuch Shale before it lithified to a state that limited the ionic exchange.

The secondary magnetization which we observed is not identifiable in all samples. One possible explanation lies in the closeness of some of the primary directions of magnetization to the mean pole of the secondary component as can be seen in Fig. 5. Another explanation is the permeability of the samples which display the secondary magnetization. The secondary component of magnetization is most easily identifiable in samples of the Freda which are the coarser-grained siltstone and sandstones, and unidentifiable in the shales, especially the samples of the Nonesuch. A sandstone unit which lies at the contact between the Copper Harbor Conglomerate and the Nonesuch Shale does display the secondary component, thus eliminating stratigraphic control as an explanation. If the secondary magnetization is of a chemical nature, a variable permeability of the section explains its presence or absence. Were the secondary magnetization of thermal origin, there would be no preference for the porous members. A thermal origin for the secondary magnetization cannot be eliminated entirely, but is limited to low temperatures of 100°C or less. Such temperatures are, of course, not at all uncommon, but seldom appear to result in ancient secondary magnetizations with moderately high blocking temperatures.

Conclusions

With the number of Precambrian pole positions continuously increasing, it is possible to better constrain the polar wandering path for North America. It is important to note that the clustering of poles for late Keweenawan times is not limited to the Lake Superior sampling sites (Fig. 6). The poles from the Freda Sandstone (1050 Ma; 179.5° E, 1.2° N) and the Nonesuch Shale (1060 Ma; 176.5° E, 10.3° N) agree very well with the pole determined from the Pikes Peak Granite (1030 Ma; 181° E, 6° N; Spall 1970) located in Colorado in another tectonic province of North America. The mean pole from the Portage Lake Lavas of middle Keweenawan age (1100 Ma; 184° E, 25.2° N; Books 1972; Irving and McGlynn 1976) is in similar agreement with rocks of the same age from the Gila County Diabase (1150 Ma; 179° E, 27° N; Helsley and Spall 1972) located in Arizona. When reliable poles for rocks of a given age are determined, anomalous poles such as those outside the cluster in Fig. 6 can be reexamined. This will be of increasing importance for the unraveling of the complex tectonics of Precambrian areas such as the Grenville Province, Newfoundland, and others.

Once polar wandering paths are well constrained for the Paleozoic and Precambrian times, secondary components of magnetization may be used to determine the age of the events causing the magnetic overprinting. The chemical event which we believe to be responsible for the secondary magnetization yields a pole position near late Hadrynian and Cambrian poles. This age agrees well with the best estimates of the time of emplacement of the White Pine ore body. In a similar manner, secondary components which are of thermal origin may be used to date thermal events. When secondary components of magnetization are processed to yield secondary poles, the age determination of the magnetic overprinting may make possible the study of paleochemical and paleothermal events.

The methodology of thoroughly sampling extensive stratigraphic columns of sedimentary rocks can aid in the determination of apparent plate movement. Assuming that the magnetization is acquired during deposition or shortly thereafter, observed variations in magnetic directions can be explained by plate motion or secular variation. When secular variation is smoothed out, the movement of the plate may be observed. This procedure has been carried out for the upper Keweenawan sedimentary rocks and the plate motion observed is in agreement with the general trend of the North American plate as determined by Spall (1971), Irving and McGlynn (1976), and others.

Acknowledgments

The authors thank Ray Leone from Copper Range Company and Robert Seasor from the White Pine Copper Company for their assistance in locating outcrops and acquiring maps of the area. We are indebted to our colleagues Henry N. Pollack and William C. Kelly (University of Michigan), and Dennis V. Kent (Lamont-Doherty Geological Observatory) for their helpful discussions and critical review of the manuscript, and Mark L. Grummon (University of Michigan) for his guidance in the wet chemical analysis. We are also indebted to the reviewers, J. Roy and H. C. Halls, for several constructive criticisms. Fieldwork was made possible while working on another project in the area under the financial support of the Advanced Research Projects Agency contract number F 44620-73-C-0060, whereas the laboratory studies were partly funded by grant GA 38038 from the National Science Foundation.

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